

The encoder, not the bins: decomposing peak-level MS/MS gains

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Abstract

Machine-learning methods for tandem mass spectrometry (MS/MS) have been moving away from fixed-width binned intensity vectors towards models that read peaks at full precision, and the resulting gains are commonly attributed to the spectral representation. That attribution has never been tested, because abandoning bins normally forces an architecture change at the same time: a peak set cannot be read by the network that reads a binned vector. Here we separate them. We introduce a rounding control that snaps each peak to its bin centre, merges the peaks that collide, and passes the result through the *identical* peak-set Transformer at an identical parameter count, so the encoder changes with the mass axis — the precision of the m/z scale — held constant, and the mass axis changes with the encoder held constant. On the CASMI 2026 corpus, over 8,692 held-out molecules and three training seeds, replacing a 0.5 Da binned baseline with a full-precision peak-set model raised macro AUPRC from 0.0478 to 0.1122 against a chance level of 0.0093. The control splits the paired gain of +0.0645 into +0.0377 (95% CI +0.0331 to +0.0424) for the encoder and +0.0268 (+0.0221 to +0.0317) for the mass axis: 58% of the gain is architectural. A bin-width sweep at fixed encoder shows mass precision paying only below 0.05 Da, with 0.5, 0.25 and 0.1 Da mutually indistinguishable. Improvements measured against binned baselines should therefore not be read as evidence about spectral representation. Code: <https://github.com/xulinpan/cheminformatics>.

Scientific contribution

We introduce a rounding control that isolates the spectral representation from the encoder in MS/MS substructure prediction, an isolation no previous comparison achieves because changing the representation ordinarily forces an architecture change. Applying it across three seeds shows that the encoder accounts for 58% of the gain a two-arm comparison credits to the representation and is the largest of the three effects we measure, while a bin-width sweep at fixed encoder locates the value of mass precision below 0.05 Da. The consequence for practice is that performance measured against a binned baseline confounds representation with architecture and cannot support claims about either alone.

Keywords

Tandem mass spectrometry, Structure annotation, Spectral representation, Mass defect, Substructure prediction, Ablation study

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Introduction

Assigning a chemical structure to an MS/MS spectrum is the rate-limiting step in untargeted metabolomics. Where the molecule appears in a reference library the problem reduces to spectral matching, but most spectra acquired from biological extracts do not [1]. For the remainder, computational annotation relates the spectrum to candidate structures: by in-silico fragmentation and scoring, as in MetFrag [2]; by predicting a molecular fingerprint and searching a structure database, as in CSI:FingerID [3] and MIST [4]; or by learning a spectrum representation whose geometry reflects chemical similarity, as in MS2DeepScore [5] and DreaMS [6].

For most of the past decade these systems took the same input object. The spectrum, physically a sparse list of (m/z , intensity) pairs, was discretised onto a fixed grid of bins and handed to the model as a dense vector. The convention came from spectral library search, where a bin comfortably wider than the instrument’s mass error makes cosine matching robust to calibration drift between laboratories. A 0.5 Da bin spans roughly 1,700 ppm at m/z 300 while a modern time-of-flight instrument delivers about 5 ppm, so everything the instrument resolves inside that window is discarded — including the mass defect, the difference between a fragment’s exact and nominal mass, which distinguishes C_5H_7^+ (67.0542) from $\text{C}_4\text{H}_3\text{O}^+$ (67.0178) and therefore carries elemental composition.

The field has since moved away from naive binning. CSI:FingerID scores against fragmentation trees over exact masses, MIST tokenises peaks by chemical formula, DreaMS operates on peak lists, and TransExION aligns fragments by nominal mass with the defect embedded as a separate channel [7]. These systems outperform binned baselines, and the improvement is routinely read as evidence that the representation was the problem.

That reading does not follow from the experiments, and the reason is structural rather than careless. A binned vector and a peak set are different kinds of object: the first is dense and fixed-length, the second sparse and variable-length and permutation-invariant. No architecture reads both. Abandoning bins therefore forces a change of encoder, and every published comparison of a peak-level model against a binned baseline varies the representation and the architecture together, usually alongside differences in training data and objective. The two contributions have never been separated, so the field’s estimate of what the representation is worth rests on experiments that cannot supply one.

This paper supplies the separation. The obstacle above is real but not insurmountable: while a peak-set encoder cannot read a binned vector, it can read peaks whose m/z has been *rounded to the bin grid*. Quantising the mass axis — we use the term throughout for the precision of the m/z scale, the component of the input representation that binning actually destroys — while leaving the peak-set interface intact yields an arm that carries the baseline’s mass information through the treatment’s architecture, and inserting it between the two turns a confounded two-arm comparison into a decomposition. We report that decomposition across three training seeds, extend it with a bin-width sweep that prices precision as a continuous quantity, and examine what the results imply for how baselines in this area should be constructed and read.

Methods

Corpus and splits

We used the CASMI 2026 training corpus, an aggregation of eleven public and in-house MS/MS

libraries. All experiments use a fixed subset of two of its twenty-one Parquet row groups: 262,041 spectra over 43,536 distinct structures, comprising 15,299,995 peaks after cleaning. Spectra were retained if the precursor m/z was below 2,000 and the reported precursor error within 20 ppm; peaks were retained above 0.1% of the base peak and below the precursor plus 1.5 Da, capped at the 256 most intense, with m/z held in double precision throughout preprocessing.

Structures were assigned to five folds by grouping generic Murcko scaffolds, with acyclic structures falling back to grouping by their own identifier, and fold sizes balanced greedily (8,707 to 8,708 molecules). Folds 2–4 were used for fitting, fold 1 for model selection, and fold 0 was held out; all reported numbers are on fold 0 (8,692 molecules).

Structural targets

Each structure was decomposed with BRICS and represented by the presence or absence of each of the 512 most frequent attachment-labelled fragments, the vocabulary fitted on the training folds alone. The target matrix is 97.1% zero, with a median of 17 positive targets per molecule. On the held-out fold 477 targets carry at least five positives, at a mean prevalence of 0.0093; that prevalence is the chance level for macro AUPRC. Because 475 of 600 candidate targets fall below 1% prevalence, we report AUPRC as the primary metric and AUROC only as a secondary summary.

The four arms

The design is a ladder in which each arm differs from the one below it in exactly one respect.

M0, the conventional baseline. Each spectrum is projected onto 2,048 fragment bins and 2,048 precursor-minus-fragment bins of width 0.5 Da over $[0, 1024)$, with square-root intensity accumulated into each bin and the concatenated vector L_2 -normalised. The 4,096-dimensional vector is encoded by a two-layer perceptron. This is the representation used, with minor variation in bin width, throughout the MS/MS machine-learning literature.

M1R, the rounding control. Each peak’s m/z is assigned to a cell by $\lfloor m/w \rfloor$ with $w = 0.5$ Da — the same indexing M0’s encoder uses — and placed at that cell’s centre. Peaks sharing a cell are merged by accumulating square-root intensity, the accumulator M0 applies, so the merged peak carries the total M0 would have seen. The resulting tokens pass through the peak-set encoder described below. The arm therefore holds mass information at M0’s resolution while using M1’s architecture.

One detail is load-bearing. The encoder derives the neutral loss as $m^{\text{prec}} - m$. Quantising the peaks while leaving the precursor at full precision would return the precursor’s mass defect to every token through that subtraction, silently defeating the control, so the precursor is snapped to the same grid whenever quantisation is active. A unit test asserts that two precursors in one cell give identical embeddings under the control and different embeddings without it.

M1, full precision. Identical to M1R with quantisation disabled, so peaks reach the encoder at the precision the instrument reported. M1R and M1 are identical in every other configuration field and have identical parameter counts (712,336); a test asserts both.

M2, hierarchical aggregation. Identical to M1 except that mean pooling over a molecule’s spectra is replaced by a measurement model weighting each spectrum by a learned, acquisition-dependent precision. This arm is included to give the other two effects a scale, not as a contribution of this paper.

Peak encoding

Each peak is encoded as a token. For a peak at m/z m in a spectrum with precursor m^{prec} we

form a multi-scale sinusoidal embedding

$$\varphi_L(m) = [\sin(2\pi m/\lambda_1), \cos(2\pi m/\lambda_1), \dots, \sin(2\pi m/\lambda_L), \cos(2\pi m/\lambda_L)]^\top, \quad (1)$$

with $L = 32$ wavelengths spaced geometrically from 2 mDa to 10^3 Da, applied both to m and to the neutral loss. Short wavelengths resolve the mass defect and long wavelengths carry nominal mass. The token also receives the explicit defect $\delta = m - \text{round}(m)$, square-root and log-transformed intensity, and intensity rank.

At a wavelength of 2 mDa and m/z 342 the phase in Eq. (1) exceeds 10^6 , and evaluating it in single precision destroys exactly the low-order information the embedding exists to preserve. We therefore compute the phase in double precision and reduce it modulo 2π before casting. With this reduction the embeddings of $\text{C}_4\text{H}_3\text{O}^+$ and C_5H_7^+ , which share a 0.5 Da bin, differ by 1.82 in the infinity norm.

Tokens are encoded by a four-block Transformer with a learned pooling token, each block conditioned on the acquisition covariates (adduct, polarity, instrument family, harmonised collision energy) by adaptive layer normalisation. The encoder is permutation invariant and accepts variable peak counts.

Bin-width sweep

To price precision as a continuous quantity rather than at one grid, we repeated the M1R construction at $w \in \{0.25, 0.1, 0.05, 0.01\}$ Da, with the encoder, parameter count, schedule and seed fixed. Together with $w = 0.5$ (M1R) and full precision (M1) this gives six points differing only in how finely the mass axis is resolved.

Training protocol

Every arm was trained for fifteen epochs with AdamW, a one-cycle schedule peaking at 10^{-3} , batches of 16 molecules, mixed precision and gradient clipping at 1.0. The loss was binary cross-entropy over the 512 presence targets with an auxiliary negative-binomial term on 88 functional-group counts, and checkpoints were selected on validation macro AUPRC. No hyperparameter search was performed for any arm. The four ladder arms were each trained three times under seeds $\{2026, 7, 13\}$; the sweep arms were trained once each.

Evaluation

We report macro AUPRC over targets with at least five held-out positives, with macro AUROC, micro AUPRC and Brier score as secondary summaries. Because arms are evaluated on identical molecules and identical targets, the appropriate contrast is paired: each target’s average precision is first averaged over the three seeds, the per-target differences are then taken, and their mean is bootstrapped over targets (5,000 resamples). We report the fraction of targets improved alongside, and, for each effect, its standard deviation across seeds — which, unlike the bootstrap interval, reflects initialisation variance. Where an ordering between effects is at issue we check it within each seed rather than reading it off the means.

Results

The four arms

The experiment is a ladder of four models in which each rung differs from the one below it in

exactly one respect (Table 1). M0 is the conventional baseline: a 0.5 Da binned intensity vector read by a two-layer perceptron. M1R keeps that binned mass axis but replaces the perceptron with the peak-set Transformer, snapping each peak to its bin centre and merging the peaks that collide, so the encoder receives tokens carrying no more mass information than the binned vector did. M1 hands the same Transformer the peaks at full precision. M2 adds hierarchical pooling over a molecule’s spectra. Data, targets, splits, optimiser, schedule and evaluation are fixed throughout; M1R and M1 are identical in every configuration field but the width of the mass grid, and carry identical parameter counts. Each arm was trained three times under seeds {2026, 7, 13}.

The rung that matters for the question is M1R. Without it, the contrast between a binned baseline and a peak-set model changes the mass axis and the encoder at the same time, and a difference between them cannot be assigned to either. M1R holds the mass axis at the baseline’s resolution while giving it the treatment arm’s encoder, so the two halves separate.

Table 1: The four arms, each differing from the one above it in a single respect. Macro AUPRC on 8,692 held-out molecules, mean over three training seeds. Chance is 0.0093 over 477 scored targets.

arm	mass axis	encoder	pooling	parameters	macro AUPRC	SD over seeds
M0	0.5 Da bins	MLP	mean	1,164,560	0.0478	0.0006
M1R	0.5 Da bins	Transformer	mean	712,336	0.0854	0.0014
M1	full precision	Transformer	mean	712,336	0.1122	0.0039
M2	full precision	Transformer	hierarchical	829,584	0.1444	0.0016

Seed spread

Across seeds the arms are stable enough to compare, but not uniformly so (Table 1). M0 varies by 0.0006 and M2 by 0.0016, while M1 varies by 0.0039 — roughly six times M0’s spread and, as it turns out, of the same order as the differences we wish to resolve. Any claim about the relative size of the three effects therefore has to be checked seed by seed rather than read off the means, and we do so below.

Most of the apparent representation gain is the encoder

Replacing the binned vector with a full-precision peak set raises macro AUPRC from 0.0478 to 0.1122, a paired per-target gain of +0.0645 (95% CI +0.0576 to +0.0713) improving 92.5% of targets — the effect one would report from a two-arm comparison. The control splits it in two (Fig. 1, Table 2). Changing the encoder alone, with the mass axis binned in both arms, is worth +0.0377 (+0.0331 to +0.0424; 90.6% of targets). Changing the mass axis alone, with the same encoder on both sides, is worth +0.0268 (+0.0221 to +0.0317; 74.8% of targets). The two components sum to the joint effect exactly, as they must, since each target’s contribution enters both sides linearly.

Fifty-eight per cent of what a two-arm comparison would attribute to the spectral representation is therefore the encoder. Both components are real: each interval excludes zero by a wide margin and each improves a clear majority of targets individually. But they are not what the headline number suggests, and the larger of the two is the one that has nothing to do with how the spectrum is represented.

What the ordering does and does not support

The encoder is the largest of the three effects, and this survives replication: it exceeds the mass-axis effect in all three seeds (by +0.0139, +0.0154 and +0.0033) and the aggregation effect in all three

Table 2: Paired per-target differences in average precision, bootstrapped over targets after averaging each target’s AP across the three seeds. The final column is the standard deviation of the effect across seeds.

contrast	isolates	Δ AUPRC	95% CI	targets improved	SD over seeds
M1R – M0	encoder	+0.0377	[+0.0331, +0.0424]	90.6%	0.0019
M1 – M1R	mass axis	+0.0268	[+0.0221, +0.0317]	74.8%	0.0049
M2 – M1	aggregation	+0.0322	[+0.0275, +0.0370]	79.5%	0.0035
M1 – M0	encoder and mass axis	+0.0645	[+0.0576, +0.0713]	92.5%	–
M2 – M0	all three	+0.0966	[+0.0873, +0.1064]	94.1%	–

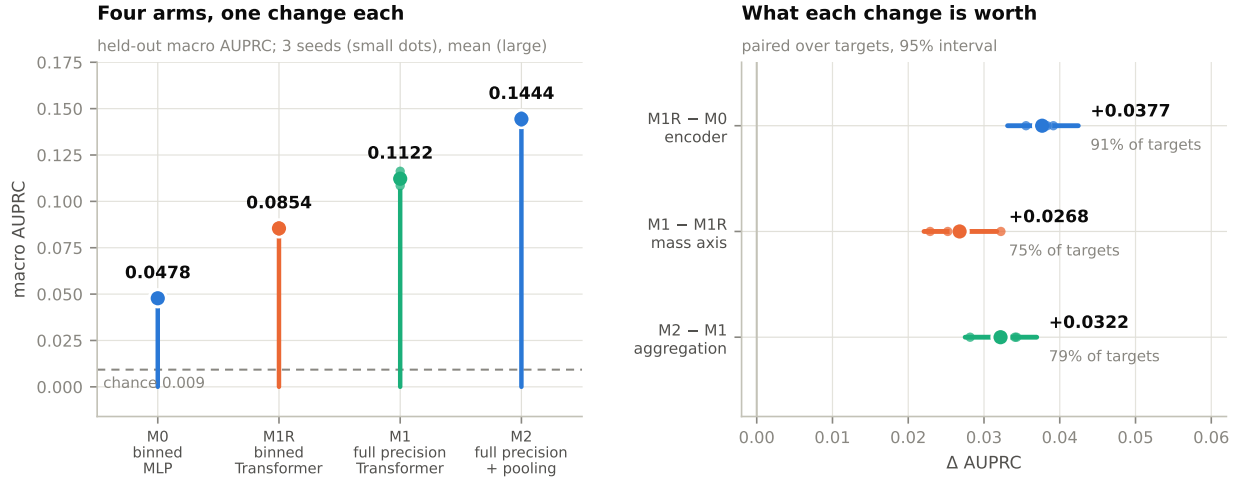


Figure 1: Decomposing the gain. **Left:** held-out macro AUPRC for the four arms; small dots are individual seeds, large dots their mean. **Right:** what each single change is worth, paired over targets with 95% bootstrap intervals; small dots are the per-seed effect sizes.

(by +0.0051, +0.0040 and +0.0074). We regard that as established at this sample size.

The ordering of the remaining two is not. The mass axis averages +0.0268 and aggregation +0.0322, which would place the representation last, but the sign of that difference reverses under one of the three seeds (−0.0089, −0.0114, +0.0041). Three seeds cannot separate effects that differ by 0.0054 when the per-seed spread of the difference is of the same magnitude. We therefore claim only that the mass axis and the pooling strategy are of comparable value, and that both are smaller than the encoder. A reader wanting the two ranked will need more replicates than we have run.

This matters for how the experiment should be read. It would be convenient to report that the spectral representation is the smallest of the three effects; the data do not support that, and the conservative statement — that it is not the largest — is the one the replication licenses.

How much precision is needed

The rounding control prices one particular grid. Sweeping the grid width, with the encoder and parameter count fixed throughout, prices precision as a function of how much of it is retained (Fig. 2, Table 3). At 0.5, 0.25 and 0.1 Da the model sits between 0.0801 and 0.0867, a range of 0.0066 — under twice the full-precision arm’s seed standard deviation of 0.0039, and unordered within itself; coarsening from 0.1 to 0.5 Da costs nothing measurable. Below that the curve moves: 0.05 Da reaches 0.0909 and 0.01 Da reaches 0.1030, closing 65% of the distance to full precision. Full precision reaches 0.1119, so even a 10 mDa grid — fine enough to separate the isobaric fragments that motivate the whole argument — leaves roughly a third of the precision benefit on the table.

Two cautions. Each width was run once, and the non-monotonic dip at 0.25 and 0.1 Da spans 0.0066, under twice the full-precision arm’s seed standard deviation of 0.0039 and therefore unremarkable across five single runs; we read it as noise rather than structure. We tested whether it might instead be an aliasing artefact — the sinusoidal basis has wavelengths that are integer divisors of these grid spacings, which would render some channels constant — but only one to three of thirty-two channels are affected at any width and the count does not track performance ($r = -0.26$), so we discount that explanation. Resolving the middle of this curve would need seeds at each width.

Table 3: Bin-width sweep, single seed. Every arm uses the M1 encoder at an identical parameter count; only the width of the mass grid changes. The final column expresses each result as a share of the distance between the 0.5 Da arm and full precision.

bin width (Da)	peaks absorbed	macro AUPRC	share of the 0.5 Da gap closed
0.5	37.5%	0.0867	0%
0.25	36.8%	0.0818	-19%
0.1	33.1%	0.0801	-26%
0.05	25.9%	0.0909	17%
0.01	7.5%	0.1030	65%
full precision	0.0%	0.1119	100%

What binning removes

The mechanism is visible before any model is fitted (Fig. 3). Of the 93,611 evaluation peaks that survive preprocessing and are seen by every arm, 35,116 — 37.5% — fall into a 0.5 Da bin already occupied by another peak from the same spectrum. The loss falls with the grid: 33.1% at 0.1 Da, 25.9% at 0.05 Da, 7.5% at 0.01 Da. The observed mass defect spans $[-0.47, +0.47]$ Da with clear

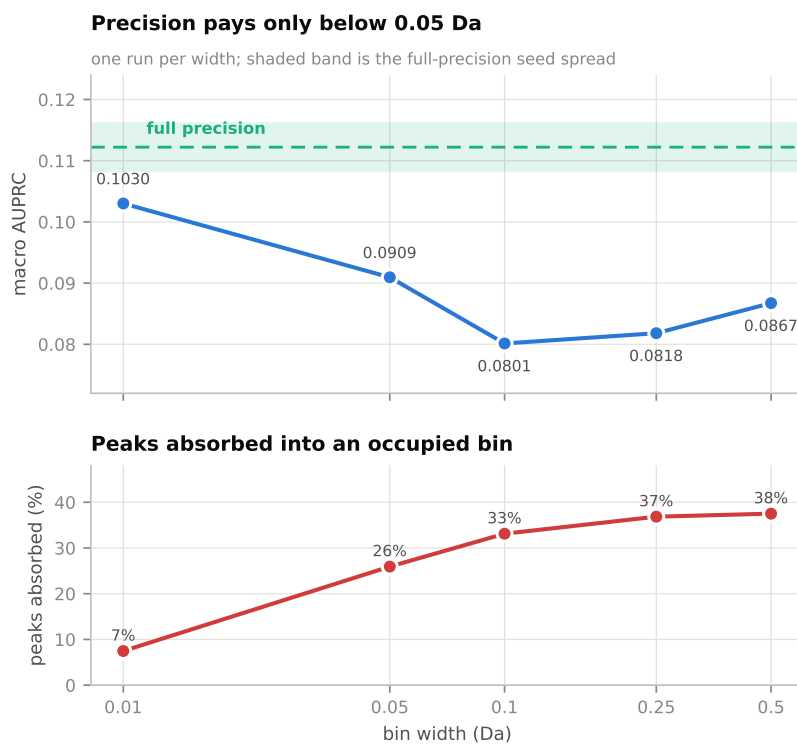


Figure 2: Bin-width sweep. **Top:** macro AUPRC against grid width, one run per width; the band is the full-precision arm's seed spread. **Bottom:** the fraction of peaks absorbed into an already-occupied bin at each width.

compositional structure, and enumerating plausible CHNOS compositions within 10 ppm places 328 candidate fragment formulas inside a single 0.5 Da bin at m/z 300.

The denominator deserves a note, because the obvious one is misleading. The raw test files contain 400,707 peaks, of which 60.7% collide under 0.5 Da binning. Preprocessing removes 76.6% of those as sub-threshold before any arm is trained, and the removed peaks are disproportionately the dense low-intensity ones that collide. Quoting the raw figure would charge binning for peaks no model ever receives, so we report absorption over the cleaned spectra throughout.

Comparing the absorption curve with the performance curve is instructive, and slightly deflationary. Going from 0.5 to 0.1 Da removes about a tenth of the absorption and buys nothing measurable; going from 0.05 to 0.01 Da removes about two thirds of the remainder and buys +0.0121. Peak absorption is therefore not a sufficient account of the cost on its own: what the model gains below 0.05 Da tracks the recovery of the defect value itself more closely than it tracks the recovery of peak count.

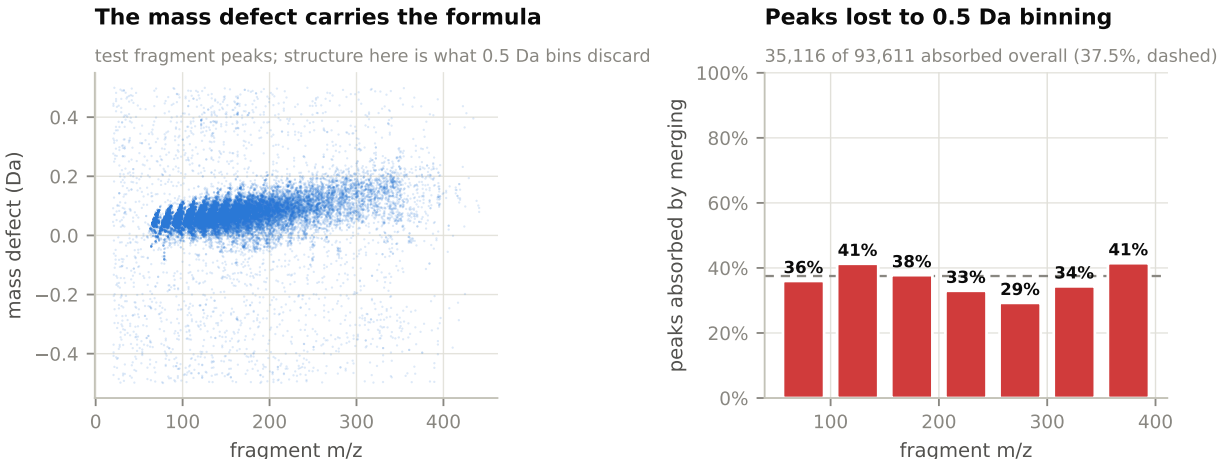


Figure 3: Information discarded by 0.5 Da binning, over the cleaned evaluation spectra. **Left:** mass defect against fragment m/z . **Right:** fraction of peaks absorbed into an already-occupied bin, by mass range; the dashed line marks the overall 37.5%.

Discussion

The headline is a reattribution. A two-arm comparison on this corpus credits the spectral representation with +0.0644 macro AUPRC; once the encoder is held constant, the representation’s share is +0.0268 and the encoder’s is +0.0377. Both are real, and the smaller one is the one the literature has been discussing.

It is worth being clear about why this went unnoticed for so long, because the answer is not inattention. The confound is forced by the data structures involved. A binned vector is dense and fixed-length; a peak set is sparse, variable-length and permutation-invariant. There is no architecture that reads both, so an experiment that changes only the representation appears impossible, and the comparison everyone can run is the one that changes both. The rounding control works not by finding a network that reads both objects but by degrading one object into the other while keeping the interface fixed — passing peaks through the peak-set encoder at the baseline’s resolution.

The manoeuvre is simple in retrospect, and we suspect it generalises: wherever a representation change forces an architecture change, asking whether the richer representation can be impoverished in place is likely to be more productive than looking for a universal encoder.

Our result does not diminish the systems that abandoned binning. CSI:FingerID, MIST, DreaMS and TransExION are better than binned baselines, and this work does not dispute it. What it changes is the account of *why*. To the extent that those systems differ from a binned baseline in both respects, our decomposition suggests the larger part of their advantage comes from the architectures they adopted rather than from the mass information they preserved. That is a testable claim about each of them, and the control we describe is cheap enough to apply.

The sweep sharpens the picture and, in one respect, complicates it. Precision buys nothing measurable between 0.5 and 0.1 Da; the gain appears below 0.05 Da and is not exhausted even at 0.01 Da. A 10 mDa grid separates the isobaric fragments that motivate the mass-defect argument, yet still leaves about a third of the benefit unrealised, which suggests the model is using the defect as a continuous quantity rather than merely as a discriminator between candidate formulas. This sits awkwardly with the common design choice, TransExION’s included, of embedding the defect as a channel alongside a coarsely quantised nominal mass: on our evidence the resolution of that channel matters more than its presence. The comparison of the absorption and performance curves points the same way. Coarsening from 0.1 to 0.5 Da removes a tenth of the peaks by merging and costs nothing; refining from 0.05 to 0.01 Da removes two thirds of the remaining merges and buys +0.0121. Peak absorption, the most intuitive account of what binning destroys, is therefore not sufficient on its own.

The practical consequence concerns controls. A method evaluated against a binned baseline is being compared against an arm that is handicapped in two ways at once, and on this evidence the larger handicap is architectural. A paper reporting a gain over such a baseline has not shown that its representation is responsible, whatever its discussion says, and reviewers should not accept that inference. The remedy is inexpensive: report the comparator’s encoder as explicitly as its representation, and where the claim is about representation, include an arm that changes only that.

Limitations

The experiments use approximately a tenth of the available corpus, and absolute values will move with scale even if the ordering does not. Three seeds establish that the encoder effect is the largest and that all three components are non-zero, but not enough to rank the mass axis against aggregation, and we have declined to do so. The sweep is single-seed, so only its endpoints should be read as findings. Finally, and most substantively, M1R does not separate dense from sparse encoding: a Transformer over bin-indexed tokens is not a perceptron over a dense vector, and some of what we attribute to the encoder may be that switch rather than the attention mechanism. A fifth arm — a perceptron over the rounded peak set, or a Transformer over bin tokens without merging — would resolve it, and we regard that as the natural next experiment rather than a caveat that undermines the present one.

Conclusion

Binned intensity vectors entered MS/MS machine learning from spectral library search, and the field’s move away from them has been read as evidence that the spectral representation was the

limiting factor. That reading was never tested, because no architecture reads both a binned vector and a peak set, so representation and encoder have always changed together. Inserting an arm that carries binned mass information through a peak-set encoder separates them. On 8,692 held-out molecules across three seeds, 58% of the gain attributed to the representation is the encoder (+0.0377) and 42% is the mass axis (+0.0268), and a sweep at fixed encoder locates the value of precision below 0.05 Da. Mass precision matters, but less than the architecture adopted alongside it, and improvements measured against binned baselines cannot be used as evidence about either.

Supplementary Information

Additional file 1 reports per-target average precision for every arm and seed, the full bootstrap distributions, the per-seed effect sizes, and the peak-absorption analysis broken down by adduct and collision energy.

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Author contributions

X.P. designed the study, implemented the models and the control, performed the experiments and analyses, and wrote the manuscript. C.W. contributed to the interpretation of the chemical results and revised the manuscript. Both authors read and approved the final manuscript.

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Availability of data and materials

All code, configurations, trained-run reports and the analysis that generates every number in this paper are available at <https://github.com/xulinpan/cheminformatics> under the MIT licence. The CASMI 2026 corpus is distributed by the competition organisers.

Declarations

Competing interests

The authors declare no competing interests.

References

- [1] Hoffmann MA, Nothias L-F, Ludwig M, Fleischauer M, Gentry EC, Witting M, Dorrestein PC, Dührkop K, Böcker S (2022) High-confidence structural annotation of metabolites absent from spectral libraries. *Nat Biotechnol* 40:411–421

- [2] Ruttkies C, Schymanski EL, Wolf S, Hollender J, Neumann S (2016) MetFrag relaunched: incorporating strategies beyond in silico fragmentation. *J Cheminform* 8:3
- [3] Dührkop K, Shen H, Meusel M, Rousu J, Böcker S (2015) Searching molecular structure databases with tandem mass spectra using CSI:FingerID. *Proc Natl Acad Sci USA* 112:12580–12585
- [4] Goldman S, Wohlgend J, Stražar M, Haroush G, Xavier RJ, Coley CW (2023) Annotating metabolite mass spectra with domain-inspired chemical formula transformers. *Nat Mach Intell* 5:965–979
- [5] Huber F, van der Burg S, van der Hooft JJJ, Ridder L (2021) MS2DeepScore: a novel deep learning similarity measure to compare tandem mass spectra. *J Cheminform* 13:84
- [6] Bushuiev R, Bushuiev A, Samusevich R, Brungs C, Sivic J, Pluskal T (2025) Self-supervised learning of molecular representations from millions of tandem mass spectra using DreaMS. *Nat Biotechnol*
- [7] Bui-Thi D, Liu Y, Lippens JL, Laukens K, De Vijlder T (2024) TransExION: a transformer based explainable similarity metric for comparing IONS in tandem mass spectrometry. *J Cheminform* 16:61